Behavioral Biometrics

Handprint Biometrics

Behavioral Biometrics

- Behavioral biometrics identifies individuals based on **behavioral patterns**.
- Examples:
  - Keystroke dynamics
  - Signature dynamics
  - Gait recognition
  - Hand movement patterns

Handprint Biometrics

Handprint biometrics combines:

- Physiological features (hand shape)
- Behavioral traits (pressure, movement pattern)

- Identification using the shape, geometry, and dynamic characteristics of the human hand during interaction.
- It may include:
  - Hand geometry
  - Palm shape
  - Finger length and width
  - Pressure distribution
  - Writing or touch dynamics

Introduction to Handprint Biometrics

What is Handprint Biometrics?

Handprint biometrics uses the unique geometric, textural, vascular, and dynamic characteristics of the human hand for automated identity recognition. First commercialised in the 1980s for physical access control, it remains one of the most widely deployed biometric modalities — with Fujitsu PalmSecure alone installed in 3,500+ hospitals globally.

Hand Geometry

Measures physical shape: finger lengths, widths, inter-finger spacing, palm length and width. Simple 17–25D feature vector. Low cost, fast, widely deployed in access control.

Palm Print

Captures palm surface texture — principal lines, wrinkles, ridge minutiae. High-res version overlaps with fingerprint features. 150–500 DPI images required.

Palm Vein Pattern

Near-infrared (850 nm) imaging reveals subcutaneous vein topology. Contact-less, highly hygienic, extremely difficult to forge. ~5 million unique feature points.

Behavioural Hand

Captures dynamic parameters: grip force, contact pressure distribution, touch-screen stroke patterns, pen-hold dynamics, and interaction timing over time.

Types of Handprint Biometrics

1. Static Handprint (Physiological)

- Hand geometry
- Palm dimensions
- Finger measurements

2. Dynamic Handprint (Behavioral)

- Pressure variation
- Motion trajectory
- Speed of hand placement
- Touchscreen interaction patterns

Types of Hand Biometric Features

Geometric Features

- Finger lengths (F1–F4, thumb) measured at each phalanx segment
- Finger widths at proximal, medial, and distal joints
- Palm length: wrist crease to base of middle finger
- Palm width: across the metacarpal bone line
- Inter-finger spacing at knuckle and fingertip level
- Aspect ratios and angular relationships between digits

Palm Print Features

- Three principal lines: heart line, head line, life line
- Secondary wrinkles — fine texture descriptors
- Ridge minutiae: bifurcations and ridge endings
- Ridge orientation and frequency maps (like fingerprint)
- Texture features: SIFT, ORB, LBP, Gabor wavelets
- Requires 150–500 DPI sensor for full minutiae capture

Vein / NIR Features

- Captured at 850 nm near-infrared wavelength
- Haemoglobin absorbs NIR — vascular paths appear dark
- Bifurcation and endpoint topology of the vein network
- Graph-based or image-correlation matching used
- Completely internal — unaffected by surface skin state
- Unique per finger, per hand, and between twins

Behavioural / Dynamic

- Grip force profile: how tightly and where hand squeezes
- Contact pressure distribution map across palm surface
- Area of contact evolution: how grip changes over time
- Pen-hold angle, stroke velocity, and altitude/azimuth
- Touch-screen swipe curvature, speed, and micro-tremor
- Temporal rhythm: characteristic inter-event timings

System Architecture

- Data Acquisition
- Preprocessing
- Feature Extraction
- Template Generation
- Matching
- Decision

Data Acquisition

Captured using:

- Digital camera
- Flatbed scanner
- Infrared sensor
- Touchscreen with pressure sensors
- Depth cameras (3D hand capture)

In dynamic systems:

- Motion sensors
- Pressure-sensitive pads

Preprocessing

- Noise removal
- Background segmentation
- Hand region extraction (After segmentation, Contour detection)
- Edge detection (Sobel operator, Canny edge detector, Prewitt operator)
  - (Measuring finger length, Detecting palm width, Extracting geometric features)
- Image normalization (scaling, rotation, Alignment using reference points)

Goal:
Standardize hand images before feature extraction.

Measured Anatomical Features

Feature	Description
F1 – Finger Lengths	Proximal, medial & distal phalanx chain
F2 – Finger Widths	Measured at 3 joint positions each
F3 – Palm Length	Wrist crease → base of middle finger
F4 – Palm Width	Inner to outer metacarpal edge
F5 – Knuckle Geometry	X,Y coords of 12 knuckle landmarks
F6 – Inter-finger Spacing	Distance between adjacent finger bases
F7 – Thumb Geometry	Length, width, angle vs index finger
F8 – Principal Lines	Heart, head, life line on palm surface

Feature Extraction (Static Features)

Common static features:

- Hand length
- Palm width
- Finger length
- Finger thickness
- Area of palm
- Geometric ratios

Mathematical representation:

$F=[f_1,f_2,f_3,\ldots,f_n]$ Where each f_i is a measurable hand feature.

Feature Extraction (Dynamic Features)

Behavioral features include:

- Pressure variation over time
- Speed of hand placement
- Angle of placement
- Touch sequence patterns
- Motion trajectory

Time-based feature vector:

$$F(t)=[p(t),v(t),\theta(t)]$$

Where:

$p(t)$ = pressure
$v(t)$ = velocity
$\theta(t)$ = angle

Feature Extraction Techniques

Geometric Measurement
Classical

- 17–25D vector: finger lengths/widths, palm dims, inter-finger gaps
- Fast Euclidean or city-block distance matching
- Low cost, interpretable, real-time on embedded MCU
- Typical FAR ~0.1%; best for low-security access control

PCA — Eigenhand
Statistical

- Flatten normalised palm image → high-dimensional vector
- PCA computes eigenpalms (principal eigenvectors)
- Project test image; compare in reduced eigenspace
- Compact template; sensitive to illumination variation

Gabor Filter Bank
Texture

- Bank of Gabor wavelets: 6–8 orientations × 3–5 scales
- Captures crease, ridge orientation and frequency
- Phase response quantised → PalmCode (binary template)
- Hamming distance matching: microsecond speed

Local Binary Patterns (LBP)
Texture

- LBP code per pixel: threshold vs. 8 neighbourhood pixels
- Build histogram over grid of palm sub-regions
- Concatenate histograms → final feature vector
- Rotation-invariant; excellent on low-resolution scans

SIFT / ORB Keypoints
Keypoint

- Scale-/rotation-invariant interest points on palm
- 128-D SIFT or 256-bit ORB descriptor per keypoint
- FLANN / Hamming distance matching
- Handles partial occlusion and viewpoint variation

Deep CNN (ArcFace / ViT)
Deep Learning

- Fine-tune ResNet-50 / EfficientNet on CASIA-PalmprintV1
- 512–2048D embedding from penultimate FC layer
- ArcFace / CosFace metric loss for discriminative space
- State-of-the-art: >99.5% rank-1 on PolyU dataset

Template Generation

Features converted into:

- Numerical vector
- Time-series model (for dynamic traits)

Template stored in database:

User ID	Feature Vector
101	[x1, x2, x3...]

Compact representation improves matching efficiency.

Matching Techniques

Static Matching:

- Euclidean Distance
- Manhattan Distance
- Correlation-based similarity

Dynamic Matching:

- Dynamic Time Warping (DTW)
- Hidden Markov Models (HMM)
- Neural Networks

Decision:

If similarity score > threshold → Match

Matching & Classification

Distance & Similarity Metrics

Euclidean Distance

$$d = \sqrt{\Sigma(x_i - y_i)^2}$$

Geometric feature vectors; fast for low-dimensional data

Hamming Distance

$$HD = (A \text{ XOR } B)/N$$

Binary PalmCode / GaborCode — microsecond comparison speed

Cosine Similarity

$$sim = A \cdot B \,/\, |A||B|$$

CNN embedding vectors — scale-invariant angular comparison

City-Block (L1)

$$d = \Sigma|x_i - y_i|$$

LBP histogram matching — robust to feature outliers

Classification Methods

k-Nearest Neighbour

1-NN with Euclidean distance — baseline for hand geometry. Fast for small enrolment databases.

Support Vector Machine

RBF-kernel SVM on geometric or texture features. Excellent for 2-class verification tasks.

Random Forest

Ensemble of 100–500 decision trees. Handles mixed feature types; resistant to overfitting.

Siamese CNN Network

Twin CNNs sharing weights; trained with contrastive or triplet loss. Best for few-shot enrolment.

ArcFace / CosFace Loss

Additive angular margin loss for deep metric learning. State-of-the-art on palm print identification.

09 | Performance Metrics & Evaluation Standards

FAR — False Accept Rate

~0.01–0.1%

Palm vein achieves <0.01%

FRR — False Reject Rate

~0.1–1%

Higher in low-quality images

EER — Equal Error Rate

~0.5–1%

Threshold where FAR = FRR

CRR — Correct Recog. Rate

>99%

Rank-1 ID rate on PolyU DB

Evaluation Metrics Explained

ROC Curve

Receiver Operating Characteristic: plots TAR (True Accept Rate) vs FAR at all threshold values. Area Under Curve (AUC) should be ≥ 0.999 for high-security use.

DET Curve

Detection Error Trade-off: FRR vs FAR on log-log scale. Standard evaluation format in ISO/IEC 19795 biometric testing standard.

Rank-N Accuracy

In 1:N identification mode, probability that the correct match appears in the top-N ranked results. Rank-1 is most important.

d-prime (d')

Separability measure between genuine and impostor score distributions. Higher d' indicates better discrimination. Target: d' ≥ 3.0.

Template Ageing

Measures accuracy degradation over time as hand characteristics change. Annual FRR increase should be < 0.1% per year for stable systems.

Advantages

- ✓ Non-intrusive
- ✓ Easy to acquire
- ✓ Low-cost hardware
- ✓ Suitable for access control
- ✓ Behavioral component- increases security

Limitations

- ✕ Lower uniqueness compared to iris/fingerprint
- ✕ Sensitive to injury or swelling
- ✕ Environmental effects (lighting, pressure variation)
- ✕ Limited large-scale datasets

Applications

- Physical access control systems
- Time attendance systems
- Banking authentication (touch-based devices)
- Multimodal biometric systems
- Touchscreen-based authentication

Handprint vs Palmprint vs Fingerprint

Feature	Handprint	Palmprint	Fingerprint
Geometry	Yes	Limited	No
Texture	Limited	Yes	Yes
Behavioral	Yes (dynamic)	No	No
Accuracy	Moderate	High	Very High

Security Considerations

- Template encryption
- Anti-spoofing measures
- Liveness detection
- Pressure pattern verification

Dynamic features reduce spoofing risk.